

CAN Bus Intrusion Detection Based on Deep Learning With Data Augmentation for Connected Autonomous Vehicles

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Abstract—As the de-facto standard for in-vehicle networks, the Controller Area Network (CAN) is exposed to different types of cyber-attacks due to the lack of security mechanisms. Intrusion Detection Systems (IDS) can be deployed to identify the attacks by monitoring host and network activities. However, there is little abnormal historical data that can be used to train deep learning models, resulting in data imbalance and biased trained model. Hence, we propose a prediction-based IDS framework for detecting the attacks on a CAN bus, which consists of two deep-learning models of the data augmentation module and the prediction module. Firstly, the Generative Adversarial Networks (GAN) was utilized as the data augmentation module to automatically generate high-quality attack data and balance the training set. Two networks were introduced as the prediction module, and the first one is a convolutional neural networks (CNN) that predicts correlated data of all CAN IDs, and the second one is an LSTM that predicts messages individually using times series data for each CAN ID. Furthermore, an intrusion detection equipment for the CAN bus was designed and the real vehicle test was conducted. The experimental results show that the proposed method can detect CAN attacks, with an average F1-score of 99.74% and an accuracy of 99.78%. Compared with the reference work, the F1-score of attack detection is improved by 15.25%, and also the detection time is reduced by 29.11%.

Index Terms—In-vehicle networks, CAN, IDS, CNN, LSTM, data augmentation.

I. INTRODUCTION

WITH the development of Internet of Vehicles (IoVs) and mobile communication technology, the connected autonomous vehicles (CAVs) are equipped with a variety of sensors and onboard computing devices, which contribute significant improvement to driving experience, and play a vital role in Intelligent Transportation Systems (ITS) [1]. While CAV introduces a range of comfort and safety features, all of the added components, the interaction between vehicles and external

access will become more frequently, resulting in new attack surfaces and causing security issues for in-vehicle networks [2], [3], [4], [5]. As the de-facto standard for in-vehicle networks, the Controller Area Network (CAN) has become the dominant protocol used in vehicles due to its advantages of multiple master control, high transmission speed, high reliability, and low cost [6]. However, the CAN bus is a broadcast media, and a CAN message is not intrinsically provided with security mechanisms, such as encryption and authentication. Due to the simplified structure and the lack of security insight, the CAN protocol is vulnerable to many attacks, including injection, Dos, replay, drop, and tampering attacks [7]. When a hacker has collected CAN traffic, the reverse engineering analysis techniques can be utilized to analyze the vehicle's critical driving information transmitted through CAN traffic, then the hacker can get into the CAN bus through external wireless or physical access such as OBD-II Port to gain unauthorized control of a vehicle, which would compromise the safety of driver, the passengers, and the vehicles. Koscher et al. [8] developed CarShark, a software designed to monitor and manipulate messages on the CAN bus through the OBD-II Port and then control the door locks, headlights, ECUs and other components of the vehicle. In order to safeguard CAVs, Intrusion Detection Systems (IDS) can be deployed to discover suspicious network activity of in-vehicle networks and detect attack traffic.

Deep learning techniques have been extensively employed by the intrusion detection systems in ITS due to their ability to analyze data in an independent manner without relying on any human knowledge [9], [10], [11]. However, they have limitations. Since CAN specifications are kept confidential and privately owned by manufactures, it is difficult to analyze and understand in-vehicle data. These data-driven models require massive amount of historical CAN traffic to capture the data features and gain the ability to identify attack traffic. Several in-vehicle datasets have been shared to promote vehicle security research [12], [13]. Still, there are few attack samples in these datasets to train intrusion detection systems for CAN bus, so models cannot fully learn the characteristics of attack data and be fully utilized, resulting in biased trained models that could not identify attacks well [14]. To address this issue, various data augmentation approaches have been proposed (e.g., SMOTE [15] ADASYN [16]). Although the data set is expanded to a certain extent, they are essentially simple replication and rearrangement strategies, which cannot generate substantial

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data, and easily leads to over-fitting with learning. With arrival of Generative Adversarial Networks (GAN), new methods for data augmentation have appeared. Kumar et al. [17] provided an intrusion detection system based on an XGBoost Classifier, which used Wasserstein Conditional Generative Adversarial Network (WCGAN) to generate attack data and balance the training set. Yuan et al. [18] proposed a data augmentation method named B-GAN, using a combination of GAN, LSTM and RNN. The finding showed that B-GAN can generate high-quality attack samples and improve the performance of intrusion detection. As one of the most popular models in the field of deep learning, GAN has been widely used in computer vision, natural language processing and other fields, but there are few applications in the field of in-vehicle networks security [19], [20]. Song et al. [21], [14], [22] were the first to introduce GAN into the CAN bus security. They proposed several deep learning-based methods, which detected anomalies using GAN and DCNN as classifier, respectively. Nevertheless, they did not consider the spatiotemporal correlation of CAN messages contents and the authors stated that the proposed methods still show poor performance for tampering attacks detection. CAN messages are time series data and can be broadcast via different ECUs, which means there is a correlation between the ECUs since they are designed to collaborate to perform complex in-vehicle tasks. Therefore, the correlation of CAN messages can be incorporated to improve the performance of model.

The goal of this paper is to provide an IDS that can achieve high detection accuracy and low detection time in a resource limited and data imbalanced environment. Considering the temporal correlation between CAN messages, a lightweight deep learning model can be used to effectively extract spatiotemporal features. Additionally, data augmentation method can be introduced to deal with the model underfitting problem caused by the lack of attack samples. Therefore, a prediction-based IDS with data augmentation is proposed for CAN bus attacks detection. The prediction module consists of two networks, one based on CNN and the other on LSTM. The CNN module predicts correlated data for all sequential CAN IDs based on the interdependence of different ECUs. The LSTM module predicts time series data for each CAN ID using payload data that contains information crucial for the operation of the vehicle. These models leverage the correlation of CAN message contents, which makes it better suited for CAN buses with constrained resources and detection time. To address the data imbalance problem, GAN is utilized to automatically generate high-quality attack samples, making it possible to deal with the imbalance of malicious CAN traffic and achieve good performance with limited attack data. The data augmentation module generates attack data by directly processing the bit representation of the CAN ID and payload data without the need for private CAN specifications from manufacturers, which simplifies the processing procedures and makes it easier to be adapted to different manufacturers and vehicles without requiring the reverse engineering analysis techniques.

The main contributions of this paper can be summarized as follows:

- 1) We propose a novel IDS for CAN bus by fusing a CNN and LSTM with data augmentation, which is more suitable

for such a data imbalanced environment and more effective for CAN messages that carries time series physical signals of different limited-resource ECUs. The evaluation results show that the proposed method of using data augmentation to train model successfully improves the performance of the IDS, with the average accuracy is 99.74% and the average F1-score is improved by 15.25%.

- 2) Our method directly processes the bit representation of the CAN ID and payload data, which simplifies the processing procedures and makes it more efficient and effective. The experiments show that the proposed model can complete anomaly detection within 0.56 ms, which is reduced by 29.11%.
- 3) We design an equipment for CAN bus to detect and simulate online attacks on a real vehicle. The proposed model is also tested on two domain controllers. The experiments show that the accuracy and the detection time of the proposed model are close to the performance on a PC.

The rest of this paper is organized as follows. Some previous studies on intrusion detection systems for CAN bus security are reviewed in Section II. Each module of our proposed IDS structure is introduced in detail in Section III. Section IV provides the performance evaluation of the proposed method. In Section V, we introduce an equipment for CAN bus to detect and simulate attacks on a real vehicle. Finally, conclusions are provided in Section VI.

II. RELATED WORKS

The intrusion detection systems for CAN bus security have been widely investigated by several pioneer researchers over the years. Based on the detection scope and basis, the studies can be grouped into statistical and machine learning-based IDS. The statistical-based studies learn normal behavior based on conditional statistical relationship analysis and establish thresholds for anomalies detection through baseline patterns. Gmiden et al. [23] introduced an anomaly detection system using the time intervals of CAN messages to detect anomalies like Dos and replay attacks. The system checked the arrive time of each CAN messages then detected attacks based on the time intervals compared to last messages. Taylor et al. [24] provided an intrusion detection method based on the analysis of CAN data flow, which can compare the statistical value of the current and historical traffic in 1-s time window. Muter et al. [25] firstly introduced entropy into CAN bus security. The entropy of CAN messages was calculated to distinguish the Dos and replay attacks. Similarly, Wu et al. [26] also proposed an improved entropy-based IDS. They calculated the information entropy of CAN traffic over a fixed-length sliding window to detect attacks. The findings showed that the performance of their method demonstrated better than previous entropy-based IDS. Koyama et al. [27] presented a frequency-based anomaly detection method to discover violations in the network traffic using quantized intervals of the CAN payload values. HAN et al. [28] developed an IDS using survival analysis model to monitor CAN traffic and the survival rate of each CAN ID is calculated to detect anomalies. Evaluation results showed that the normal and

attack traffic could be detected with high accuracy. While these statistics-based methods showed decent performance for high-volume attacks, they did not consider the low-volume attacks that have little effect on CAN traffic flow [12].

Furthermore, several machine learning-based solutions have been proposed to detect attacks on the in-vehicle network. Avatefipour et al. [29] proposed an IDS based on a modified one-class SVM (OCSVM) in the CAN traffic. The result proved that the proposed method achieved the highest performance for anomaly detection compared to classical one-class SVM and Isolation Forest. Duan et al. [30] proposed an improved isolation forest method for tampering attacks detection. They quantified the degree of anomalies by calculating the data mass of CAN messages, and identified attacks by the anomaly score. The method was evaluated on three datasets - one generated by the CANOE and two standard datasets. It was compared to iForest, LOF, and one-class SVM and achieved an accuracy of 99.93%, which is better than the baseline. Kang et al. [31] provided an IDS based on Deep Neural Networks (DNN) to learn normal traffic and to detect anomalies. Unsupervised deep belief networks were used to initialize the parameters of DNN to improve the detection performance. Tests were conducted with a simulated dataset generated by CANOE and the result demonstrated that the performance of their method was better compared to models based on SVM and ANN with an accuracy of 98%. Considering the temporal correlation of CAN messages, several studies incorporated recurrent neural networks (RNN) with memory in their structure into their methods. Taylor et al. [32] introduced an IDS for CAN bus which uses a Long Short-Term Memory (LSTM) unit in its structure. Anomalies on the CAN messages can be individually detected for each CAN ID without the need to extract features from CAN messages during the preprocessing stage. Zhang et al. [33] proposed an anomaly detection system for CAN bus by fusing CNN and RNN with an attention mechanism. The temporal features of each ECU can be automatically learned without professional knowledge. Qin et al. [34] introduced GAN-based intrusion detection system named GPIDS, to detect attacks in CAN traffic by fusing a generative adversarial network and assisted contextual pattern-aware. They achieved an average accuracy of 97.42% by performing this experiment using dataset performed on four vehicles that had six types of attacks: bus-off, spoofing, masquerade, replay, fuzzy and SOME. Jeong et al. [35] proposed an IDS for in-vehicle network which uses raw payloads to improve the intrusion detection performance. They introduce several modules to reach the trade-off between prediction performance and the computational consumption. Gao et al. [36] designed a CAN image generation method to convert CAN traffic data into CAN images, which makes the attack visible and traceable. The experiments show that the proposed IDS has a decent performance for Dos attack detection. However, their method did not consider the data imbalance issues. Therefore, it may not be as effective at distinguishing between normal traffic and previously unseen attacks. Guo et al. [37] [38] proposed an attack detection scheme for connected vehicles by fusing a refined kinematic model, measurement model, and attack model.

The experiment results show that their methods demonstrate decent detection performance and high reliability.

III. METHOD

The ability of most algorithms to identify and detect intrusions relies on historical data. Therefore, in order to establish a reliable intrusion detection model, a large amount of attack data of CAN traffic is required during the training stage. However, attack data is difficult to be obtained in the real world, and when deep neural network is used, data imbalance may cause overfitting and improper bias problems. In fact, it is difficult to perform the known attacks for all IDs inside a vehicle. Due to the safety issues, it is difficult to collect attack data while driving a vehicle. Furthermore, there is no standard specifying the meaning of IDs and encoding of the signals transmitted in the payload data, which means that manufacturers can design their own policies for different vehicles. Even for the same manufacturer, it may use different combinations of IDs and payloads due to the types of vehicle models. Additionally, CAN messages of a vehicle differs due to different driving scenarios, driver habits, road conditions and so forth. Thus, the objectives of this paper are to solve the data imbalance problem and achieve high detection accuracy in the case of resource-constrained and extremely limited attack samples, making it easier to be adapted to different vehicles from various manufacturers without requiring the reverse engineering analysis techniques and the private CAN specifications from manufacturers.

To accomplish this, a method is proposed, which involves a data augmentation module based on WGAN-GP to generate high-quantity attack samples, and a prediction module based on lightweight models to identify anomalies using the spatiotemporal correlation of CAN messages. The complete structure of our proposed framework is depicted in Fig. 1. The training samples consisting of CAN ID and its payload are converted from 1D time series vector to 2D image. Then, the data augmentation module trains WGAN-GP architecture to learn the distribution of the normal CAN traffic data. At this point, the generator of WGAN-GP is used to generate substantial CAN attack samples and balance the training set. Actually, the occurrence of normal data is significantly more than that of attack data, and then, it is suggested to collect as much normal data as possible for grasping the main feature. So, the WGAN-GP can be trained with normal data to generate attack data. With the increase of normal data in the training process, the real permitted range for each signal will become clearer. Finally, the prediction module consists of a LSTM and a CNN, which is trained with the augmented dataset to distinguish between normal traffic and attack data by using the spatiotemporal correlation of CAN messages contents.

A. Feature Extraction

In the feature extraction stage, the messages from the CAN bus are collected, then the ID and payload of each CAN message is extracted and arranged by timestamp. Furthermore, the payload is categorized based on ID. The features selected from CAN messages are taken as input for subsequent detection. These

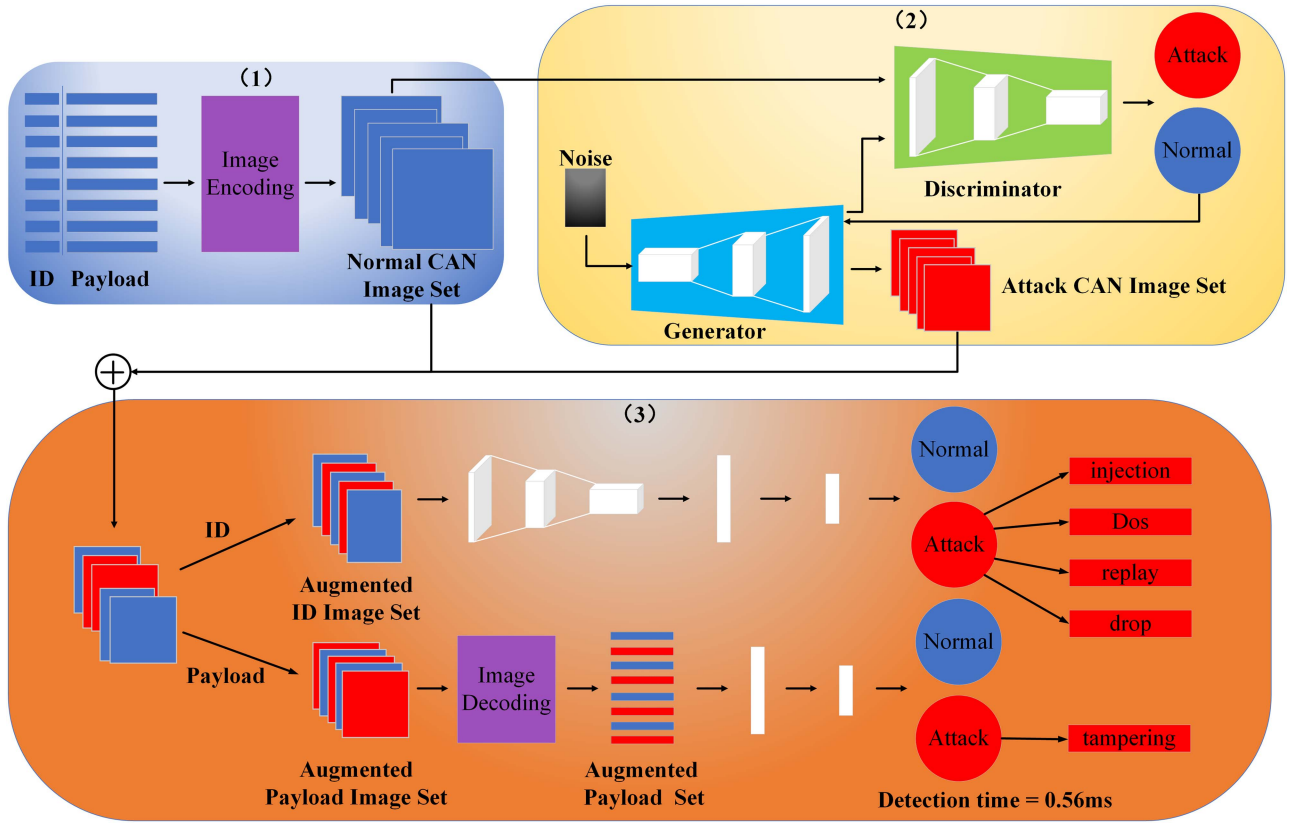


Fig. 1. The algorithm structure of the proposed IDS.

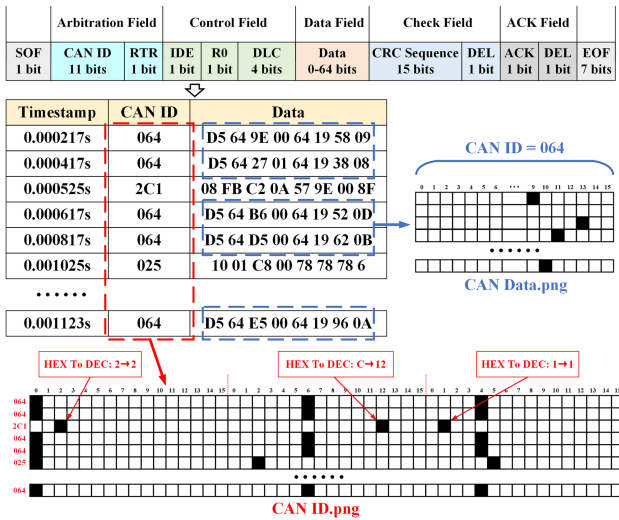


Fig. 2. The process of converting CAN messages into images.

features include the CAN IDs and eight bytes of payload data. Since GAN is introduced to generated attack samples for data augmentation and GAN has advantages in image processing, the 1D time series vector of the selected features is mapped into the 2D image with one-hot-vector. The process of converting CAN IDs and payloads into images is illustrated in Fig. 2. The 11-bit CAN ID is mapped into [1,48] matrix and the 64-bit payload into a [1,256] matrix. Specifically, each element of

CAN ID and payload is converted from hexadecimal string to binary from with 16 digits using one-hot-vector encoding. For instance, each element of the CAN ID '0x2C1' is represented by a 16-dimensional vector with 15 bits are '0' and only one bit is '1'. Then, each CAN ID is represented as a [1,48] matrix while the payload is represented as a [1,256] matrix. After that, 64 successive CAN IDs are spliced to generate a CAN ID image, and successive 256 payloads of each ID are spliced to generate a CAN payload image. In the model training stage, both IDs and payloads are converted into images and input into WGAN-GP to generate attack samples for data augmentation, and LSTM-CNN is trained using the augmentation datasets for intrusion detection. In the model testing stage, IDs are converted into images and input into the CNN to predict the correlated data of all IDs. A window is set to buffer the w most recent frames, and the frames in a window are sorted in chronological order. When the window initialization is completed, the IDS only needs to use the latest CAN message to update the window to come to a conclusion. The payloads of each message are directly input into the LSTM to predicts messages individually using times series data for each ID.

B. GAN-Based Data Augmentation for Anomaly Detection

The WGAN-GP is trained to generate synthetic images of anomalies CAN traffic and balance the training set of intrusion detection system. GAN is a generative deep modelling approach designed based on game theory, which has been recently used

for data augmentation [39]. A GAN architecture consists of the generator and the discriminator, which corresponds to a minimax two-player game. The training stage updates parameters and optimizes objective function through the competition between the generator and the discriminator.

The generator takes noise samples as input to generate fake samples. The discriminator takes the real data and the fake samples created by the generator as input to distinguish real data and fake samples.

Wasserstein Generative Adversarial Network (WGAN), an enhanced version of GAN, was introduced to solve the vanishing gradient problem of GAN [40]. Wasserstein distance W is introduced to measure the difference between the distribution of real data and fake samples, which can be defined as:

$$W(P_r, P_g) \approx L = \min_G \max_D P(D, G) \quad (1)$$

$$= E_{x \sim P_r(x)} [D(x)] - E_{z \sim P_g(z)} [D(G(z))]$$

where D is the discriminator, $D(x)$ represents the probability of an anomaly in real samples x , G is the generator, $G(z)$ represents fake samples generated from random noise z , P_r represents the distribution of real data, and P_g is the distribution of fake samples. The discriminator is required to satisfy the 1st order Lipschitz constraint, we will have:

$$\nabla_{\bar{x}} D(\bar{x}) \leq I \quad (2)$$

where $\bar{x}$ represents the linear interpolation between real samples x and random noise z , which can be calculated as:

$$\bar{x} = tx + (1 - t)z, \quad t \in [0, 1] \quad (3)$$

However, WGAN has limitations in the time consumption and the model convergence. To address this issue, the Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP), was introduced by Ishaan et al. [41]. WGAN-GP introduces a gradient penalty method to compel the discriminator to satisfy the 1st order Lipschitz function constraint. Optimal engineering results are achieved by constraining gradient values around 1, and thus, the gradient penalty term GP is stated as:

$$GP \triangleq E_{x \sim P(x)} \left[(\|\nabla_x D(\bar{x})\|_2 - 1)^2 \right] \quad (4)$$

The optimization objective of the discriminator is to maximize the probability of correctly distinguishing between fake samples $G(z)$ and real samples x , which can be defined as:

$$\max_{\theta} L(D, G) = E_{x \sim P_r(x)} [D(x)] - E_{z \sim P_g(z)} [D(G(z))]$$

$$- \lambda E_{x \sim P(x)} \left[(\|\nabla_x D(\bar{x})\|_2 - 1)^2 \right] \quad (5)$$

where λ is a hyperparameter used to adjust the gradient penalty term.

The optimization objective for the generator is to minimize the Wasserstein distance between the distribution of generated samples P_g and the distribution of real samples P_r . Due to $E_{z \sim P_g} [D(G(z))]$ is independent of the generator, the objective function for the generator can be defined as:

$$\min_{\phi} L(D, G) = -E_{z \sim P_g} [D(G(z))] \quad (6)$$

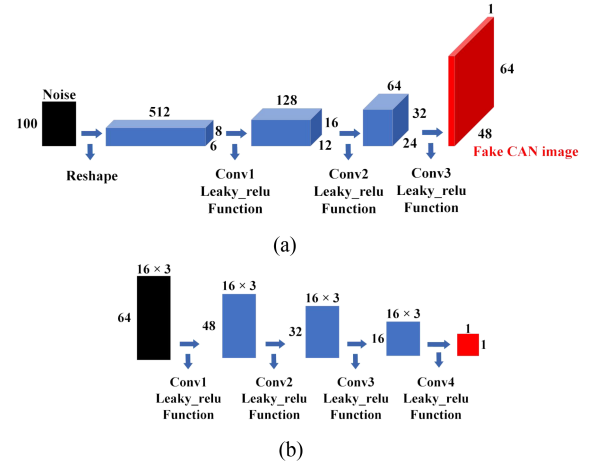


Fig. 3. Architecture of WGAN-GP based on CAN ID. (a) Architecture of generator. (b) Architecture of discriminator.

The data augmentation module consists of two WGAN-GP models. Since each ECU collaborates to perform various tasks in the vehicle, there is a correlation between ECUs. It means that payloads with different IDs are temporally correlated at a specific point in time. For example, a vehicle's speed, engine power, torque and emission values are all affected when it detects an obstacle ahead and then brakes. In addition, payloads with the same ID is time series data and tend to change with a certain periodicity. Each field of the payload is divided by different signals, and the data of each field is transmitted continuously and changes regularly within a specific range. For instance, ID = 0x064 represents engine data, which contains six signals in eight bytes, namely IdelRunning, EngTemp, PetroLevel, EngForce, EngPower and EngSpeed. The time interval between messages with ID = 0x064 is 10 ms, and when EngSpeed changes, EngPower and EngForce will change accordingly. The first model based on CAN ID is designed to learn the interdependence of different ECUs, and the second based on CAN payload is designed to learn the temporal correlation of time series data. Both of them consist of a discriminator and a generator. The only difference between them is the parameters of the network architecture. As can be seen in Figs. 3(a) and 4(a), the generator consists of a deconvolutional neural network composed of four layers, which takes random noise as the input to generate fake CAN images that are similar to the real CAN images. The discriminator, as depicted in Figs. 3(b) and 4(b) consists of a deep neural network composed of four layers, which is used to identify the fake CAN images that generated by the generator. Moreover, an early stopping mechanism is employed while the training of the intrusion detection model to avoid overfitting.

C. Prediction Module Neural Network Design

Two lightweight networks are trained to identify anomalies from normal CAN traffic, based on the spatiotemporal correlation of CAN messages contents.

1) *CNN-Based Network*: The CAN messages are correlated due to different ECUs collaborate to perform complex in-vehicle

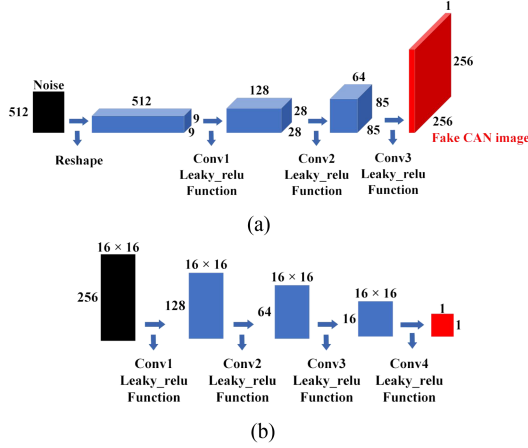


Fig. 4. Architecture of WGAN-GP based on CAN payload. (a) Architecture of generator. (b) Architecture of discriminator.

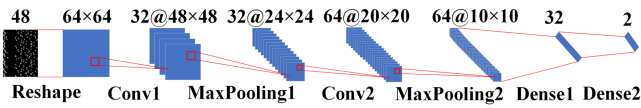


Fig. 5. CNN-based prediction network architecture.

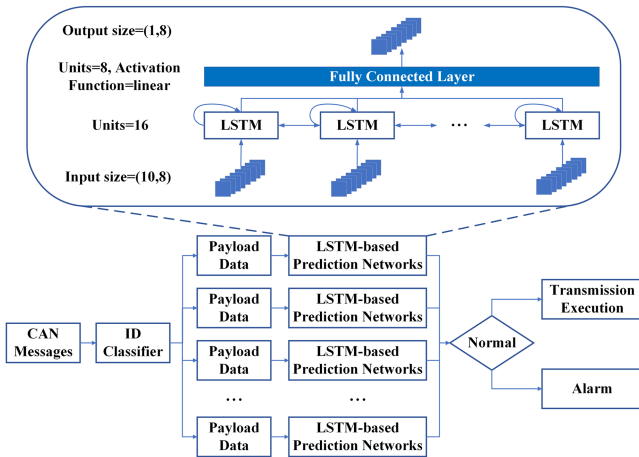


Fig. 6. LSTM-based prediction module architecture.

tasks. Therefore, we use a lightweight CNN architecture here to learn the spatial contiguity in normal CAN traffic. As can be seen in Fig. 5, the CNN prediction network, which consists six layers, four convolutional layers, two fully connected layers to extract CAN images. The ReLU is used as the activation function after each convolution layer and the model is trained with Adam optimizer. The CNN is trained to distinguish between attacks and normal CAN traffic, based on the CAN traffic images form.

2) *LSTM-Based Network*: Since the LSTM model is more effective to predict time series data, we convert the CAN payload images onto ID feature vectors using one-hot-vector. As can be seen in Fig. 6, the lightweight LSTM prediction network, which consists two layers, one LSTM layer and one fully connected layer to extract CAN messages. The LSTM is trained to determines whether the current CAN message has been tampered

using the temporal correlation of time series physical signals. Also, in our case, the CAN payload data is classified based on their CAN ID and then detected by the corresponding IDS. Once one of the IDSs detects anomalies, the vehicle will immediately receive an abnormal alarm.

IV. SIMULATION EXPERIMENT

The experiments are performed to compare the performance of our proposed method and other models, including LSTM-AE, OCSVM, iForest. Also, we analyze the effectiveness of the data augmentation module on prediction performance and compare our proposed method with other data augmentation algorithms, including GAN, SMOTE and ADASYN. Additionally, the quality of generated samples is analyzed using feature maps and statistics.

A. Implementation Details

1) *Experiment Environment*: The experiment is run on the equipment of AMD Ryzen 5 5600H with Radeon Graphics @3.30 GHz, equipped with NVIDIA GeForce RTX3050, and Windows 10 for the operating system.

2) *Attack Model*: The communication mechanism of CAN bus is broadcast. Each ECU on a CAN bus can receive and send messages to control the vehicle. A CAN messages is not protected with authentication mechanism. The attackers can access the CAN bus through OBD-II port or wireless. Attack messages will cause normal messages to be blocked or tampered with. When attack messages carry high priority IDs and are sent at a higher frequency than normal, the bus will be overloaded and the transmission of normal messages will be blocked. In addition, when the payload content of normal messages is tampered with, the tampered messages will cause false dashboard displays and incorrect vehicle feedback operations. A detailed description of these attacks models is as follows:

- *Injection*: The injection attack relies on the legitimate ID to be effectively implemented. The attacker injects attack messages with random IDs and payloads into the CAN bus. The ECU will feedback when the injection ID is legitimate. Subsequently, the legitimate ID can be used to generate effective attacks to interfere with vehicle operations.
- *Dos*: Vulnerabilities in the CAN bus arbitration mechanism are used to block the transmission of normal messages. Messages with the highest priority ID = 0x000 are injected into the CAN bus at frequency higher than the normal message. Since illegal messages always occupy CAN bus resources, other ECUs cannot transmit legal messages.
- *Replay*: Legitimate messages are collected and injected into the CAN bus at random times. The ID and payload are valid, but they do not comply with the temporal correlation between ECUs.
- *Drop*: In an in-vehicle network, the gateway is responsible for connecting high-speed CAN and low-speed CAN to transmit messages. An attacker can compromise and control the gateway through external wireless or physical access such as OBD-II Port. Then, the attacker can choose to discard the messages of key ECUs, such as engine ECU,

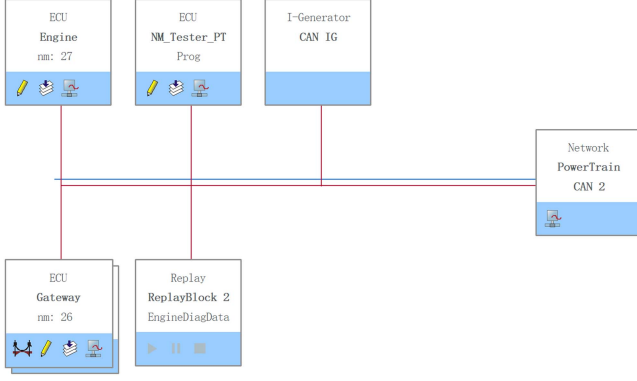


Fig. 7. In-vehicle network simulation based on the CANOE.

battery management ECU and tire pressure monitoring ECU, thus affecting the normal operation of the vehicle [20].

- **Tampering:** In this attack, the meaning of the CAN message is exposed to the attacker. The attacker can send an attack message containing a specific ID and payload by manipulating a specific ECU. As a legitimate ECU periodically transmits CAN messages with the same ID, the attack will make the data receiving node confused, unable to identify the legitimacy of the message. If the data receiving node receives a tampered message and responds, the behavior of vehicle may become abnormal.

3) **Dataset:** The dataset comes from two parts: one is the automotive controller area network bus intrusion dataset provided by Dupont et al. [42] containing real vehicle data of two cars (i.e., Renault Clio and Opel Astra) and abnormal data under five attack scenarios of injection, DoS, replay, drop and tampering attacks, the other is the simulation data generated by the CANOE (A CAN bus analysis tool developed by Vector) simulation scenario. The CAN data packets are simulated and collected by the CANOE, which consist of time sequences, ID, and payload data with eight bytes. As shown in Fig. 7, there are three ECUs in the CANOE simulation environment built in this article, including engine, gateway, and NM_Tester_PT, which transmit messages with 500 kbps CAN traffic. The IG module can customize the message type and time interval, and simulate the ECU sending messages to the CAN bus. The replay module can show the saved CAN history messages. In the preprocessing stage, we collect the original CAN traffic and group them into 12 different IDs, representing different ECUs and nodes. The payload data of each CAN ID carries information crucial for the operation of the vehicle. Table I illustrates the characteristics of different IDs in detail. To achieve the best validation loss, tree-structured Parzen estimator is utilized to automatically choose the configuration of hyper-parameter. The ranges of the hyper-parameter are given in Table II.

B. Image Encoding Size

In the feature extraction stage, the CAN IDs and payload are converted into images with widths of 48 and 256, respectively, the height of these images remains undetermined. Therefore,

TABLE I
CHARACTERISTICS OF DIFFERENT IDS

Name	ID	Total number of instances	Data length control
EngineData	0x064	26795	8
ABSdata	0x0C9	6698	6
NM_Engine	0x51B	893	4
DOOR_r	0x1F1	6698	1
Gateway_1	0x110	3349	3
Gateway_2	0x111	8931	8
Console_1	0x1A0	66989	4
Console_2	0x1A1	5359	2
NM_Console	0x41A	13397	4
NM_DOORright	0x41C	1339	4
NM_Gateway	0x41D	6698	4
NM_Gateway_PowerTrain	0x51A	669	4

TABLE II
HYPER-PARAMETER SEARCH SPACE FOR BOTH GAN AND CLASSIFIER

	WGAN-GP	Classifier
Batch size	[2 ⁵ , 2 ⁶ , 2 ⁷ , 2 ⁸ , 2 ⁹]	[2 ⁵ , 2 ⁶ , 2 ⁷ , 2 ⁸ , 2 ⁹]
Learning rate	[0.0001, 0.001, 0.01]	[0.0001, 0.001, 0.01]
Noise ratio	[0.1, 0.2, 0.3, 0.4, 0.5]	[0.1, 0.2, 0.3, 0.4, 0.5]
Kernel size	[2, 3, 4]	[2, 3, 4]

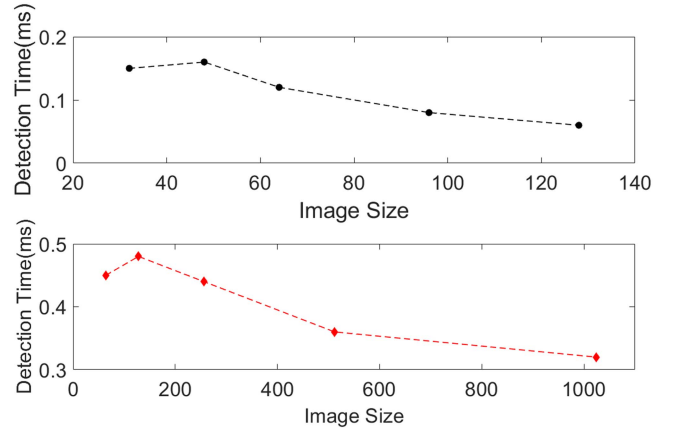


Fig. 8. Accuracy of the proposed IDS according to image size.

we analyze the relationship between the size of the CAN image and the detection performance. Specifically, the image size is increased from the minimum of 32 and 64 to the maximum of 128 and 1024, then we compare the accuracy and detection time of the proposed IDS in different image sizes. When the size of the spliced image is small, the image features are sparse and the average processing time for each message is longer, so the accuracy is lower and the detection delay is higher. As the image size increases, the average processing time for each message gradually decreases, but too large a spliced image size will also lead to model overfitting and reduce detection accuracy. As shown in Figs. 8 and 9, the results show that the best performance is achieved when the sizes of CAN image are set to be 48×64 and 256×256 , respectively.

TABLE III
STATISTICS COMPARISON

Features	Normal Data				Attack Data				Generated Data			
	Min	Max	Mean	Variance	Min	Max	Mean	Variance	Min	Max	Mean	Variance
Byte1	213	213	213	0	213	213	213	0	213	213	213	0
Byte2	100	100	100	0	100	100	100	0	100	100	100	0
Byte3	0	255	149.39	7497.89	0	255	181.77	3192.10	5	255	183.48	3544.07
Byte4	0	1	0.29	0.21	0	1	0.19	0.32	0	1	0.04	0.45
Byte5	100	100	100	0	100	100	100	0	100	100	100	0
Byte6	25	25	25	0	25	25	25	0	24	25	24.98	0.11
Byte7	0	250	123.34	5317.51	0	250	145.13	5362.27	1	245	125.62	5782.10
Byte8	8	16	10.75	5.17	8	16	11.65	8.49	0	15	9.80	9.36

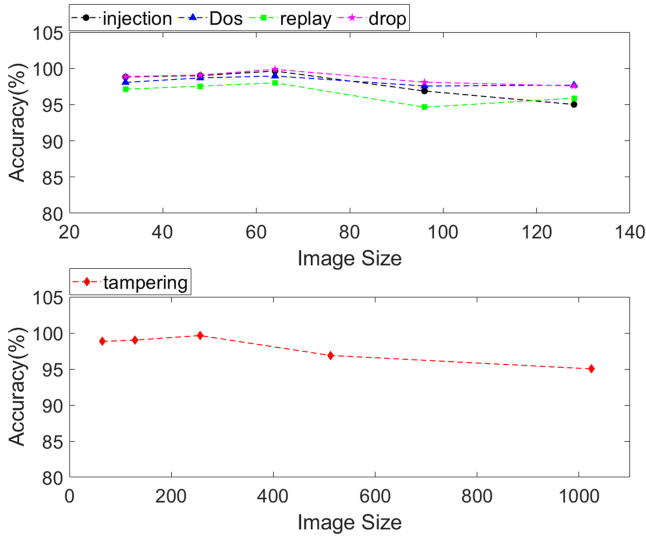


Fig. 9. Detection time of the proposed IDS according to image size.

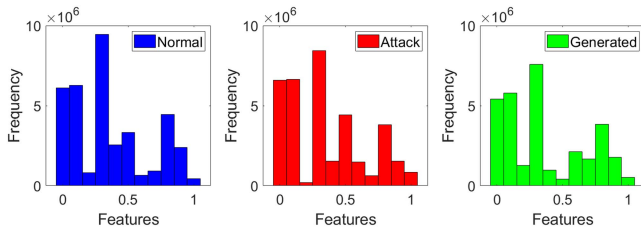


Fig. 10. The feature distribution map of normal data, attack data and real data.

C. Generated Data Analysis

To verify the effectiveness of the generated attacks, we use the feature map to compare the distribution of normal data, attack data and generated data. Due to the different scales of each feature, the data is normalized using the min-max technique, and then the frequency of the data features is counted. It can be seen from Fig. 10 that the generated data has the probability structure of the attack data, which is similar but not the same as the normal data. Also, we evaluate the effectiveness of the generated data using the range of each signal by comparing the minimum, maximum, mean, and variance of the features. In our case, we select ID = 0x064 engine data as an example.

Table III shows the range of each byte of the three data, where the EngPower signal is represented by the 1st and 2nd bytes, the EngForce signal is represented by the 3rd and 4th bytes, the PetroLevel signal is represented by the 5th byte, the IdelRunning signal and the EngTemp signal are represented by the 6th byte, the EngSpeed signal is represented by the 7th and 8th bytes. It can be seen that the range of each signal of the generated data and the attack data is close. The generated data conform to the attack data format, and each signal value is within the real permitted range. In addition, the mean difference between normal data, attack data and generated data is close to 0, which means that the byte range overlap between the three data is high. Furthermore, the difference in variance is large, which means there are still differences between the three data. The experimental results verify the effectiveness of the generated attacks and show that the generated attacks are accurate representations of actual attack.

D. Data Augmentation Analysis

1) *Varying Balance Size*: In order to analyze the contributions of the data augmentation module to the prediction performance, we vary the balance size of the training data. Specifically, the percentage of attack data is increased from 0% (original dataset) to 10%, 20%, 30%, 40% and 50%. The accuracy of our proposed model, measured by varying the training balance size in training set. As can be seen in Fig. 11, the proposed method utilizes the data augmentation module to gain accuracy. We note that the performance of all types of attacks improves as the percentage of attacks data increases within the training process. The performance of an intrusion detection model for CAN bus can be improved with accuracy achieved is lower than that on the percentage of attacks of 30%, but it is still higher than the accuracy on the original dataset (balance size = 0%) when no data augmentation has been performed. The results prove that the detection performance is successfully improved by the data augmentation module, enabling the model to detect previously unseen attacks.

2) *WGAN vs GAN vs SMOTE and ADASYS*: Also, we analyze the effectiveness of the data augmentation module by comparing it with the representative data augmentation algorithms — GAN, SMOTE and ADASYN. To do so, we train the proposed model with three different data augmentation modules, which

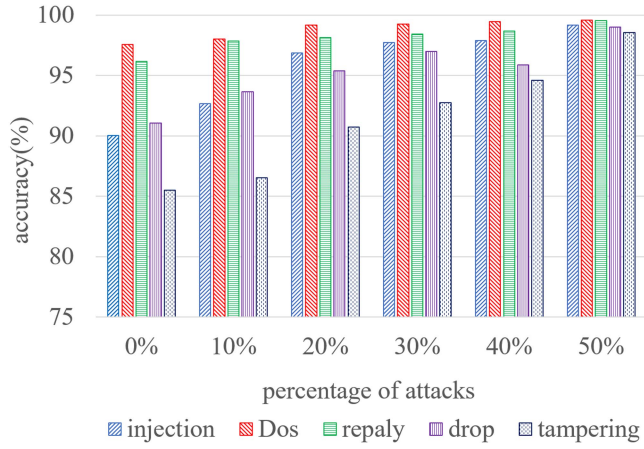


Fig. 11. Accuracy performance (axis Y), measured by varying the size of the balance (axis X) – the percentage of attacks on the entire training set – between 0% (original dataset) to 10%, 20%, 30%, 40% and 50% (augmented dataset).

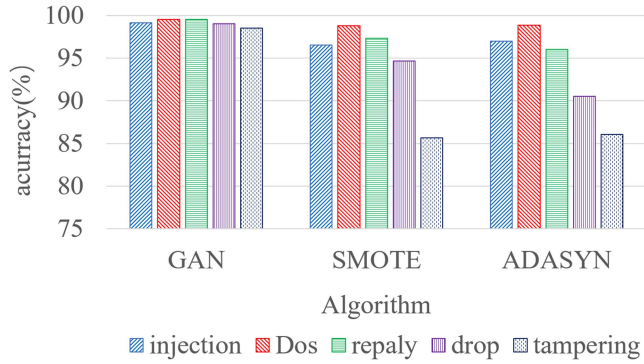


Fig. 12. Accuracy performance (axis Y), measured with WGAN, GAN, SMOTE and ADASYN on the balance size of 50%.

are denoted as GAN, SMOTE and ADASYN. The only difference between them is that the WGAN-based data augmentation is replaced with GAN, SMOTE and ADASYN, respectively. As can be seen in the Fig. 12, the WGAN-based data augmentation module provided in our proposed method perform better than the representative data augmentation algorithms in terms of accuracy, which may be related to the defects of these techniques. GAN suffers from gradient explosion problem. SMOTE based on the nearest neighbor method is easy to cause data redundancy. ADASYN based on τ distribution is susceptible to outliers.

E. Comparison to Other Algorithms

The baseline methods are representative classification methods, which are OCSVM, iForest, and a deep learning-based model called LSTM-AE. Furthermore, we compare the proposed model with three existing works (GPIDS, X-CANIDS, CanNet) discussed in Section II under five attack scenarios. We compare the detection performance of our proposed model for CAN bus with the baseline and existing works. The proposed LSTM-CNN and baseline methods are trained using public datasets and simulated datasets. Since the simulation of a CAN bus only generates normal CAN traffic, there is no attack data in the

TABLE IV
DETECTION PERFORMANCE COMPARED TO BASELINE AND EXISTING WORKS
RESULTS IN FIVE ATTACK SCENARIOS

Attack	Models	Recall	Precision	F1	Accuracy
injection	LSTM-CNN (ours)	99.9%	99.7%	99.8%	99.8%
	LSTM-CNN	62.3%	60.5%	61.4%	62.9%
	LSTM-AE (ours)	97.5%	96.8%	97.2%	97.1%
	LSTM-AE	87.2%	85.6%	86.4%	86.5%
	OCSVM	49.6%	53.0%	51.3%	55.3%
	iForest	87.3%	82.5%	84.8%	84.9%
	GPIDS [36]	98.8%	96.3%	97.5%	97.6%
	X-CANIDS [37]	99.9%	99.8%	99.9%	N/A
Dos	CanNet [38]	N/A	99.4%	N/A	99.6%
	LSTM-CNN (ours)	99.9%	99.8%	99.9%	99.9%
	LSTM-CNN	72.0%	76.3%	74.1%	73.7%
	LSTM-AE (ours)	98.8%	96.1%	97.4%	97.4%
	LSTM-AE	62.1%	66.5%	64.2%	63.6%
	OCSVM	71.4%	53.3%	61.1%	59.6%
	iForest	93.1%	60.5%	73.3%	71.2%
	GPIDS [36]	96.3%	95.8%	96.1%	96.1%
replay	X-CANIDS [37]	97.2%	97.9%	96.2%	N/A
	CanNet [38]	N/A	100%	N/A	99.7%
	LSTM-CNN (ours)	99.8%	99.9%	99.8%	99.9%
	LSTM-CNN	66.8%	61.5%	64.0%	65.3%
	LSTM-AE (ours)	94.4%	95.8%	95.1%	95.4%
	LSTM-AE	55.9%	56.4%	56.2%	57.6%
	OCSVM	51.7%	52.1%	51.9%	55.0%
	iForest	42.2%	45.2%	43.7%	46.7%
drop	GPIDS [36]	98.8%	96.3%	97.9%	97.9%
	X-CANIDS [37]	99.3%	99.9%	99.6%	N/A
	CanNet [38]	N/A	100%	N/A	100%
	LSTM-CNN (ours)	99.7%	99.6%	99.6%	99.7%
	LSTM-CNN	66.3%	63.2%	64.7%	64.8%
	LSTM-AE (ours)	85.0%	80.1%	82.5%	83.3%
	LSTM-AE	46.1%	52.2%	49.0%	49.3%
	OCSVM	55.5%	47.8%	51.4%	43.9%
tampering	iForest	61.5%	64.2%	62.8%	58.4%
	GPIDS [36]	98.5%	97.2%	97.9%	97.9%
	X-CANIDS [37]	32.9%	96.8%	40.8%	N/A
	CanNet [38]	N/A	N/A	N/A	N/A
	LSTM-CNN (ours)	99.7%	99.5%	99.6%	99.6%
	LSTM-CNN	51.8%	60.5%	55.8%	52.0%
	LSTM-AE (ours)	76.4%	84.1%	80.1%	80.9%
	LSTM-AE	42.4%	38.2%	40.2%	45.4%
	OCSVM	37.2%	45.9%	41.1%	41.4%
	iForest	31.1%	37.1%	33.8%	34.1%
	GPIDS [36]	98.7%	96.4%	97.6%	97.6%
	X-CANIDS [37]	97.2%	97.9%	96.2%	N/A
	CanNet [38]	N/A	N/A	N/A	N/A

original simulated datasets. Therefore, we use the IG module to customize the message type and time interval, and simulate the ECU sending abnormal messages to the CAN bus. In addition, the labelled LSTM-CNN and LSTM-AE methods are trained using public datasets, simulated datasets and generated datasets to evaluate the performance improvements of models trained with augmentation datasets.

The performance of these models for CAN traffic intrusion detection is shown in Table IV. It can be observed that the proposed method performs better than the baselines and existing works. We can see that the proposed method can extract the temporal correlation of CAN messages contents, and it can reach high accuracy over all types of attacks. Generally, the average accuracy of the proposed method is 99.78%, and the

TABLE V
RESOURCES CONSUMPTION COMPARED TO EXISTING WORKS RESULTS

Model	Memory footprint (KB)	Detection time (ms)
GPIDS [36]	N/A	1.97
Platform: Intel i5 CPU with twelve cores and an RTX 3060 GPU		
X-CANIDS [37]	N/A	0.79
Platform: NVIDIA Jetson AGX Xavier		
CanNet [38]	1640	1.18
Platform: Intel(R) Xeon(R) Gold 6126 CPU		
LSTM-CNN (ours)	260	0.56
Platform: AMD Ryzen 5 5600H with Radeon Graphics		

average score of F1 is 99.74%. Although the detection performance of the proposed method in injection attack is slightly lower than that of the X-CANIDS method, and the detection performance in Dos and replay attacks is slightly lower than that of the CanNet method, the accuracy of our proposed method for these attacks is still above 99.8%. Compared with the model that trained using public datasets and simulated datasets, we note that the accuracy of LSTM-CNN and LSTM-AE methods, which are trained with the balanced dataset that contains public datasets, simulated datasets and generated datasets, improved significantly by 56.54% (from 63.74% to 99.78%) and 50.17% (from 60.48% to 90.82%), respectively. The result proves that the data augmentation module implemented in our proposed model successfully improve the detection performance for CAN bus by generating high-quality attack samples. OCSVM and iForest, which are trained without generated data, show poor performance for CAN bus intrusion detection. Though the detection performance of the proposed method for tampering attacks is the worst among all types of attacks scenarios, it is still better than the other methods in Table IV. In future, the detection of tampering attacks will be a more challenging issue. Before tampering attacks occurs, the attacker listens and sniffs packet on a CAN bus. Reverse engineering technique is then used to identify the corresponding messages of each ECU. The attacker manages to override messages sent to brakes, gears, and engine and displayed fake speedometer readings on the dashboard. Even so, the accuracy of our proposed model in detecting tampering attacks is still higher than the baselines and above 99 percent.

F. Efficiency Evaluation

To evaluate whether our proposed method is suitable for CAN bus in real vehicles, we compare the proposed model with three existing works based on the resource consumption. Table V show the comparison of the proposed intrusion detection model with existing works in terms of memory footprint and average detection time per CAN frame. It can be observed that the footprint of our proposed LSTM-CNN model is smaller than GPIDS, X-CANIDS and CanNet, and the proposed model still has less detection time. Compared with the X-CANIDS, the representative model proposed above, the F1-score of our proposed model is improved by 15.25%, and the detection time of our proposed model is reduced by 29.11%. As a lightweight model, our proposed intrusion detection system for CAN bus can efficiently detect all types of attacks with better detection

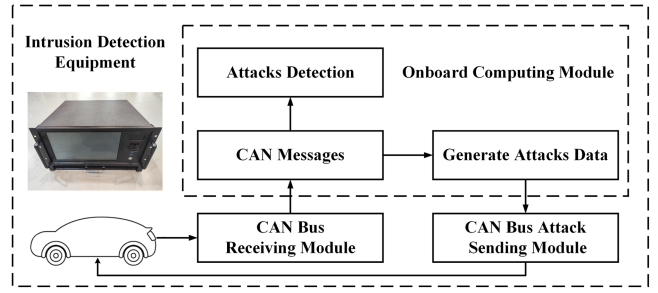


Fig. 13. The hardware architecture of the proposed equipment.

performance. In the simulation experiment, the waiting time for collecting frames is not factored in the detection time since the datasets are offline. The experimental results show that the detection time of our proposed model is 0.56 ms, which means that 1785 physical signals of different ECUs can be predicted in one second. Therefore, while our proposed model achieves a high accuracy-score, it also only consumes lower memory and less detection time, so the proposed IDS is better suited for in-vehicle networks with constrained resources and detection time.

V. INTRUSION DETECTION EQUIPMENT

A. Hardware Architecture

The proposed equipment is designed to be deployed in a gateway or an onboard unit that can receive all CAN messages. The detection performance of our proposed equipment can be used as a baseline to evaluate the detection performance of vehicles. The architecture of Windows system and microcomputer hardware is typically adopted by such testing equipment, which aligns with our proposed equipment. The size of the equipment is 430 mm × 400 mm × 215 mm. Fig. 13 shows the complete hardware architecture, which consists a CAN bus receiving module, a CAN bus attack sending module, and an onboard computing module. The main function of this equipment is to detect attacks and achieve simulation on a real CAN bus. Two micro controllers of STM32F103s are used to establish communication between the equipment and the CAN bus. The first microcontroller is used as the CAN bus receiving module to automatically match the vehicle CAN bus rate and receive CAN messages. The second one is used as the CAN bus attack sending module, which is responsible for injecting attack messages generated by the onboard computing module into the vehicle CAN bus, or inputting instructions to filter specified ID messages into the CAN bus receiving module to achieve shielded transmission of specific ECU messages. A microcomputer is introduced as the onboard computing module, equipped with Intel(R) Core (TM) i3-9100H @ 3.10 GHz, which stores the proposed model and the visualization software. The onboard computing module is responsible for intrusion detection according to the received CAN messages and generating specified types of attack messages. One end of the two STM32F103 is connected to the onboard computing module via the UART (Universal Asynchronous Receiver/Transmitter), and the other end is directly

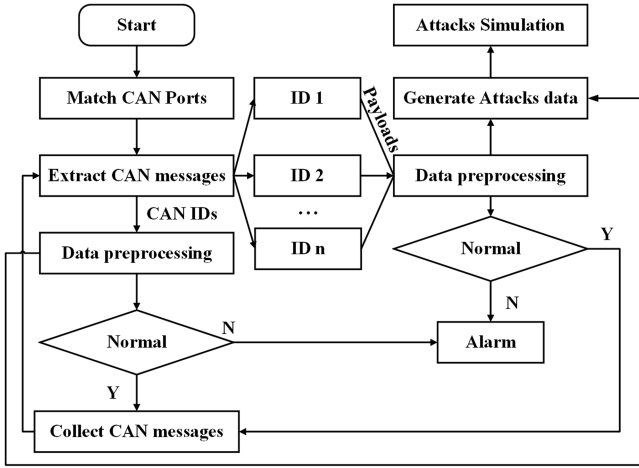


Fig. 14. The software architecture of the proposed equipment.

connected to the vehicle OBD-II port. The proposed equipment involves two communication stages. The communication baud rate between the vehicle and the CAN bus receiving module is 921600. The UART baud rate between the CAN bus receiving module and the onboard computing module is also 921600. In the vehicle testing scenario, attack messages are injected into the in-vehicle network via the CAN bus attack sending module to test whether the vehicle and the proposed method can accurately identify the attacks. The detection performance of the vehicle is evaluated by comparing the test results. This equipment is only involved in intrusion detection and does not participate in the training of neural networks. In the experiments of this article, the training of the model network relies on the AMD-based PC. Currently, this type of equipment includes detection prototypes and onboard real-time detection devices. The proposed equipment is a detection prototype. In order to facilitate experimental operation and debugging, and display CAN traffic flow, a 13 inches display screen is added, and a software interface is designed to visualize the entire algorithm process. These are also the differences compared to existing equipment. However, more and more vehicles are incorporating CAN-FD running at speeds higher than 1 Mbps. The use of UART for relaying CAN messages can still be a communication bottleneck which limits the applicability of the current setup to standard CAN. Therefore, the proposed model is also transplanted to automotive embedded devices, and we test the performance of the proposed model on two real vehicles.

B. Software Architecture

Each module of the equipment collaborates to perform vehicle testing. The equipment performs online detection of the CAN bus network, and simulates attacks in a real vehicle. The software consists of CAN bus communication establishment, data preprocessing, attacks detection and simulation. The test steps of the software are shown in the Fig. 14. To begin with, the communication between the equipment and the CAN bus is established. Then, the messages are extracted from the CAN bus and categorized based on their CAN IDs. After that, the CAN

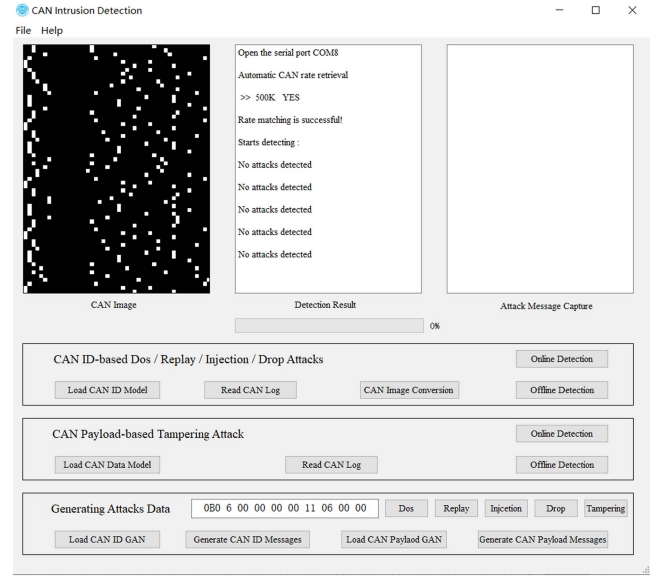


Fig. 15. The main configuration screen of the software.

IDs are converted into images and the payload of each CAN ID is processed as time series data. The IDS take these data as input for attack detection and flags anomalies once they are detected. If no attack occurs, the IDS continues to collect messages on the CAN bus and detect attacks. Additionally, the attacks data can be generated by the onboard computing module and injected into the CAN bus to simulate intrusion. The intrusion detection model development environment is python3.9.7 and tensorflow2.7.0. The software is developed based on QT and can run across platforms to facilitate Fig. 15.

C. Online Detection Evaluation for Real Vehicles

In order to evaluate the effectiveness of the proposed equipment, we performed real vehicle experiments, testing our proposed equipment on a real vehicle (RAV4). The connection between experimental components on the vehicle is shown in Fig. 16(a). As can be seen in this figure, the equipment is connected to the in-vehicle CAN networks through the OBD-II port of the real vehicle. The bandwidth of the real vehicle is automatically recognized by the CAN bus receiving module as 500 kbps. Since the tested vehicle is not equipped with an embedded intrusion detection device, we test the performance of the proposed equipment by artificially generating attacks and injecting them into the in-vehicle network. We access the CAN bus and perform all types of attacks through the CAN bus receiving module and the CAN bus attack sending module.

Due to the computing power of ECUs is differ from a PC, and intrusion detection for in-vehicle network is limited- resources. Furthermore, we test the detection performance of our model on the automotive embedded devices in two real vehicles (Volkswagen and Dongfeng). The Volkswagen is equipped with a domain controller APEX AD10, and the Dongfeng is equipped with a domain controller GEACX2, as shown in Fig. 16(b) and (c). Both the APEX AD10 and the GEACX2 are equipped with a Jetson

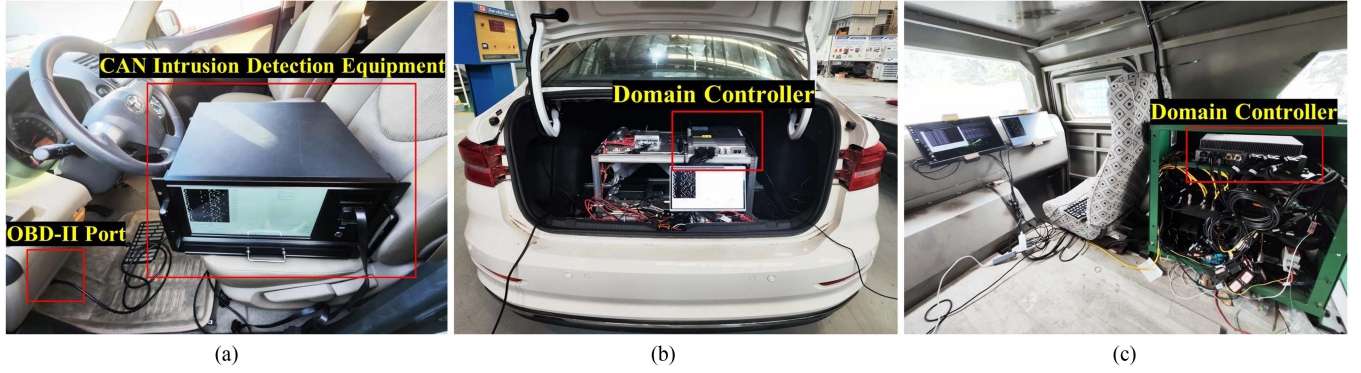


Fig. 16. Test setup on the real vehicle. (a) A RAV4 equipped with a PC. (b) A Volkswagen equipped with an APEX AD 10. (c) A Dongfeng equipped with a GEACX2.

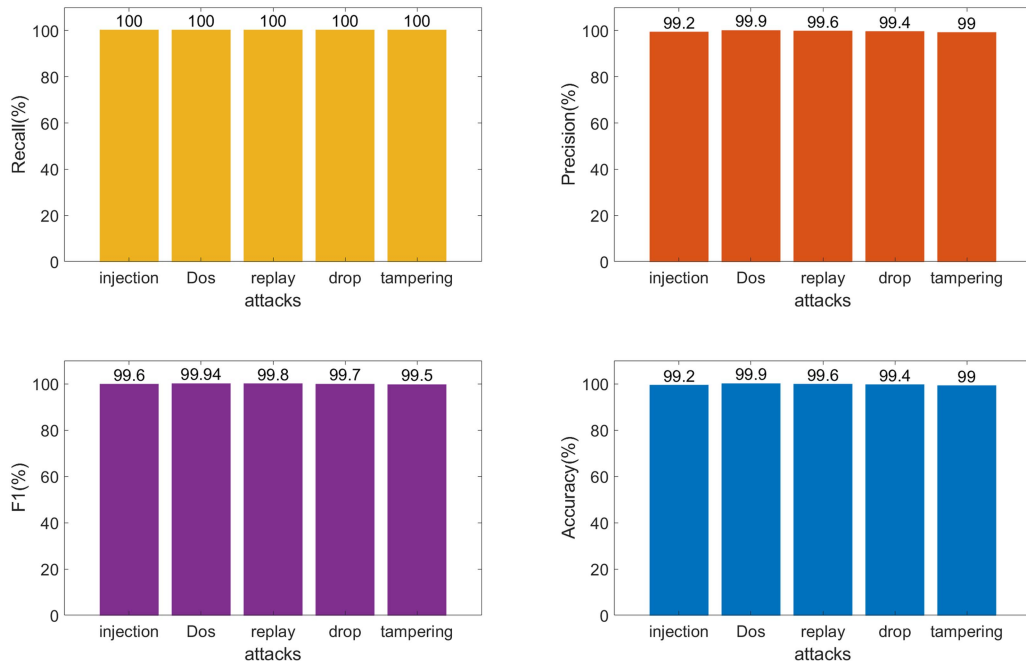


Fig. 17. Average recall, precision, F1 score and accuracy values for online detection on different platforms.

AGX Orin 32GB processor module, which includes 8 Core ARM Cortex-A78 @ 2.2 GHz and NVIDIA Ampere 1792-CUDA Core. Specifically, we cross-platform port the software containing the proposed model to the APEX AD10 and the GEACX2, respectively. Due to safety issues, we perform the experiment while the vehicle is in standby state. The CAN messages consist of different signals, including door locks, headlights, wipers, and so on. The details of attacks we simulated are illustrated as follows:

- **Injection:** CAN messages with illegal ID are injected into the CAN bus to disrupt the operations of vehicles and collect vital commands. The CAN ID is randomly generated from 0x001 to 0x7ff.
- **Dos:** CAN messages with the highest priority ID = 0x000 are injected into the CAN bus at high frequency (ie, sending period = 1 ms). The normal traffic is blocked because the abnormal traffic takes over the CAN bus.

- **Replay:** we collect normal messages and inject them into the CAN bus at random times. Due to the attack data is extracted from the normal CAN traffic, it is challenging for IDS to distinguish between authentic and fraudulent messages.
- **Drop:** we filter the CAN messages with specified ID based on the CAN bus receiving module, then, specific ECU messages are discarded.
- **Tampering:** we first collect real vehicle CAN message data, select the specified message ID by analyzing the message sending cycle, modify the data field value of the specified message, and inject it into the real vehicle CAN bus through the CAN bus attack sending module.

The average detection performance of our proposed model on different platforms is shown in Fig. 17. It can be observed that our proposed method shows good detection performance over all types of attacks. We note that the experiment results of our

TABLE VI
THE RESOURCES CONSUMPTION OF THE MODEL ON REAL VEHICLE

Platform	Memory footprint (KB)	Window Initialization Time (ms)	Frame Collection Time (ms)	Detection time (ms)
PC	406	24.65	0.39	0.82
APEX AD 10	636	26.69	0.43	1.12
GEACX2	642	26.82	0.45	1.14

proposed method achieve the best accuracy in Dos attack. The main reason is that the transmission of legitimate messages in the CAN bus is blocked by the highest priority messages sent in DoS attacks. Since DoS attacks will cause great fluctuations in CAN traffic, it is easy to detect DoS attacks. The result proves that our proposed model can achieve high detection accuracy in the case of resource-constrained and extremely limited attack samples, making it easier to be adapted to different vehicles from various manufacturers. In the real-vehicle experiment, the frame collection time is factored in the detection time. For the first message, we need to wait for window initialization to collect 63 more successive messages to come to a conclusion. When the window initialization is completed, each subsequent message can be detected by updating the window. The window initialization time and message collection time of the proposed IDS on different platforms are shown in Table VI. It can be observed that the detection time of the proposed model on PC, APEX AD10 and the GEACX2 is 0.82 ms, 1.12 ms and 1.14 ms, respectively. In addition, the average window initialization time is 26.05 ms, and the average message collection time is 0.42 ms. The experimental results show that the proposed IDS achieves high detection accuracy and low detection time in a resource limited and data imbalanced environment, which is close to the offline detection performance on a PC, so the proposed model is suitable for real vehicles.

VI. CONCLUSION

In this paper, one deep learning-based intrusion detection system framework is proposed for in-vehicle CAN bus network, including WGAN-GP data augmentation module and LSTM-CNN prediction module. CAN messages are used as input for WGAN-GP to automatically generate high-quality attack data and balance training data. The prediction module consists of two networks, one based on CNN and the other on LSTM. The CNN module is trained to detect anomalies based on the image form of the samples, which takes advantage of the spatial contiguity in CAN traffic. The LSTM module is trained to detect anomalies using the temporal correlation of time series data. The results of the experiments, performed on a simulated dataset generated through the CANOE simulation, show that the proposed method can effectively detect CAN attacks, with an average F1-score of 99.74% and an accuracy of 99.78%. Compared with the work in the related research fields, the F1-score of attack detection is improved by 15.25%, and the detection time is reduced by 29.11%. The result demonstrate that the model trained by our proposed method has greatly improved the detection performance in the case of limited resource and extremely data imbalance. Furthermore, an intrusion detection equipment is developed to detect and simulate online attacks on a

real vehicle. We also test the detection performance of our model on two domain controllers. The result proves that the proposed method is reliable and robust, making it easier to be adapted to different vehicles from various manufacturers.

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